

Muhammad Umer

■ Lahore, Pakistan | ■ +92325 8266962 | ✉ ■ muhammad.umer25052@gmail.com | www.linkedin.com/in/muhammad-umer-9ba564372 | **GitHub:** MuhammadUmer255

EDUCATION

Bachelor of Science in Software Engineering (BSSE)

2023 – 2027 (Expected)

(Currently Ongoing - Semester 5 Completed – 2.07 CGPA)

University of Central Punjab (UCP)

Lahore, Pakistan

• Relevant coursework: Data Structures & Algorithms · OOP · Database Systems · Computer Networks · Operating Systems · Software Architecture - Software Construction - Web Application Development - Advanced Web Programming

SKILLS

Languages:	Java · TypeScript · C++
Backend — Node.js:	Node.js · Express.js · REST API Design · WebSockets (Socket.IO) · JWT · Middleware Pipelines
Mobile:	React Native · Cross-platform iOS & Android
Frontend:	React.js · Tailwind CSS · HTML · CSS
Databases & Caching:	MongoDB · MySQL · Redis (caching, pub/sub, TTL) · Schema Design · Query Optimisation · Indexing
AI / Integrations:	Python ML model integration · AI/ML Enthusiat · Flask · HTTP
Tools:	Git · GitHub · Postman · VS Code · Ubuntu · Anaconda · Eclipse

NOTABLE PROJECTS

UI-Tech-Webpage

2024

HTML · CSS · JAVASCRIPT

- Developed a responsive and interactive software company website using HTML, CSS, and JavaScript, delivering a modern user interface with seamless navigation across desktop, tablet, and mobile devices.
- Designed and implemented dedicated sections for company services, technologies, projects, and contact information, improving content organization and user accessibility.
- Integrated dynamic image carousel and interactive UI components using JavaScript, enhancing user engagement and visual presentation of company offerings.
- Applied responsive design principles with CSS Flexbox, Grid, and media queries to ensure consistent performance and layout adaptability across different screen sizes.
- Optimized website structure and front-end code for improved maintainability, faster page loading, and a smooth user experience.

Smart Home Controller

2024

JAVA · JAVA SWING · GUI - OOP

- Developed a Smart Home Controller desktop application using **Java and Java Swing**, simulating a smart home environment with support for managing and controlling multiple smart devices through an interactive **GUI**.
- Built dynamic device management functionality allowing users to add, remove, and control smart devices such as Lights and Thermostats with real-time status updates.
- Designed device-specific controls including ON/OFF state management, adjustable light brightness (0–100%), and thermostat temperature settings (15°C–30°C) to emulate real-world smart home operations.
- Utilized Java Swing components and event-driven programming to create a responsive, user-friendly graphical interface and managed device collections efficiently using Array List.

Smart Parking System

2024

Java · GUI · MVC · Data Structures

- Developed a Smart Parking System desktop application using Java Swing and MVC architecture, enabling efficient management of parking operations for Cars, Bikes, and Trucks through an interactive graphical interface.
- Implemented core Data Structures including HashMap and multidimensional arrays to manage vehicle-slot mapping and parking grids, achieving O(1) average-time complexity for vehicle check-in, check-out, and slot lookup operations.
- Designed features for vehicle parking, slot reservation, reservation cancellation, and vehicle check-out with real-time parking status updates and automated slot allocation.
- Built a responsive Java Swing GUI with color-coded parking visualization, allowing users to easily monitor available, occupied, and reserved parking slots.

Social Connect

2025

REST API · MONGO DB · EXPRESS · NODE

- Developed Social Social, a full-featured social media backend API using Node.js, Express.js, and MongoDB, enabling users to create accounts, share content, interact socially, and manage personalized profiles.
- Designed and implemented RESTful API endpoints for authentication, user management, posts, stories, comments, likes, and follow functionality, supporting core social networking workflows.
- Integrated Passport.js with session-based authentication and bcrypt password hashing to provide secure user registration, login, authorization, and protected route access.
- Built a media management system using Multer, allowing users to upload and manage images for posts and stories while ensuring efficient server-side file handling.
- Modelled application data using MongoDB and Mongoose ODM, creating scalable schemas for Users, Posts, Stories, and Comments with well-defined relationships and validation rules.
- Structured the application using a modular architecture with dedicated routes, middleware, configuration, and data models, improving maintainability, scalability, and code organization.